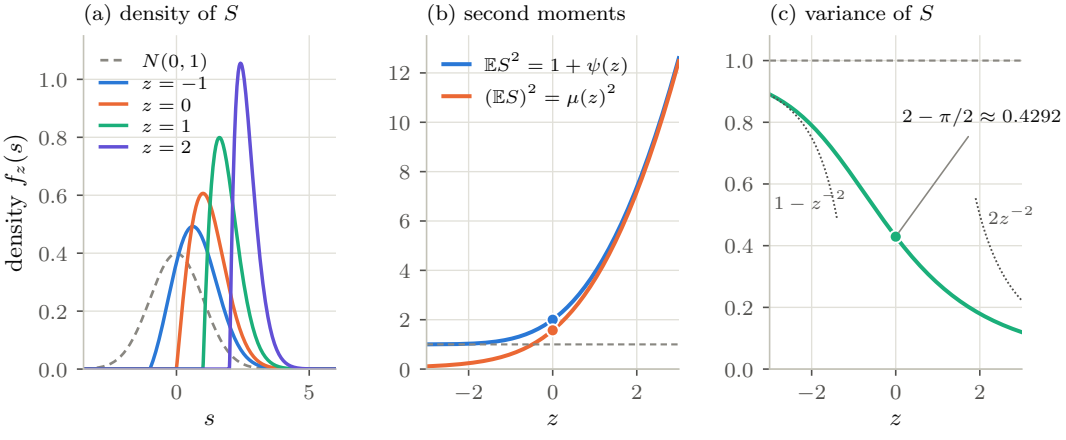

(a) density of S
$N(0,1)$
$z=-1$
$z=0$
$z=1$
$z=2$
density $f_z(s)$
s
(b) second moments
$\mathbb{E}S^2 = 1+\psi(z)$
$(\mathbb{E}S)^2 = \mu(z)^2$
z
(c) variance of S
$2-\pi/2 \approx 0.4292$
$1-z^{-2}$
$2z^{-2}$
z